

Modular Chiral Koszul Duality

The Companion

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Abstract

Classical Koszul duality is the theory of the bar construction over a point. Volume I rebuilds it over a smooth complex curve and then over the moduli of curves: the bar differential becomes an integral transform on Fulton–MacPherson compactifications with kernel $\eta = d \log(z_1 - z_2)$, its square vanishes exactly by the Arnold relation together with the Borchers identity, and at genus ≥ 1 the construction acquires curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \omega_g$ whose scalar κ is the central charge read as modular curvature. All structure assembles into one Maurer–Cartan element $\Theta_{\mathcal{A}}$ in a modular convolution Lie algebra, and the volume’s theorems -- adjunction and inversion, quadratic recognition, centre complementarity, the deformation–Hodge–graph traces, Hochschild concentration -- are its projections. The Koszul dual reflects parameters ($k \mapsto -k - 2h^\vee$, $c \mapsto 26 - c$), the conductor laws $c + c^\dagger = 2 \dim \mathfrak{g}$, 26 , $4N^3 - 2N - 2$ organise the landscape into five archetypes, and the scalar genus tower is $F_g = \kappa \lambda_g^{\text{FP}}$, the $\widehat{A}$ -genus in disguise. This companion states the engine with its exact hypothesis packages, the complete census of constants, the arithmetic shadow, and the honest frontier.

I The seed

The product of two chiral fields $a(z)b(w)$ on a smooth algebraic curve X diverges along the diagonal. Among 1-forms with a pole there, exactly one has residue independent of every choice: $\eta = d \log(z_1 - z_2)$. Three-point consistency forces the Arnold relation $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$, which generates all relations in $H^\bullet(\text{Conf}_n(\mathbb{C}))$; and η fails coordinate-invariance by the cocycle $d \log w'(z)$ -- the ghost the Virasoro algebra measures. From this seed the whole volume unfolds. The ordered bar complex of an augmented chiral algebra $\mathcal{A}$,

$$\overline{B}_X^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}) = \bigoplus_{n \geq 1} (s^{-1}\bar{\mathcal{A}})^{\otimes n} \otimes \Gamma(\overline{\text{FM}}_n^{\text{ord}}(X), \Omega_{\log}^{n-1}), \quad d = d_{\text{int}} + d_{\text{res}} + d_{\text{dR}},$$

extracts operator-product residues against the logarithmic kernel, and $d^2 = 0$ holds *if and only if* $\mathcal{A}$ satisfies the Borchers identity: configuration-space geometry selects the chiral-algebra axioms. At genus g the difference $z_1 - z_2$ has no meaning; the prime form replaces it, the propagator acquires a period correction, and the fibrewise differential is curved, $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, flatness restored only by the Gauss–Manin connection on the total space. The genus-completed differential is strictly square-zero because the Deligne–Mumford boundary satisfies $\partial^2 = 0$, and its positive part

$$\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0, \quad D\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$$

is a Maurer–Cartan element in the modular convolution dg Lie algebra -- automatic, no equation to solve. Every invariant of the volume is a projection of $\Theta_{\mathcal{A}}$: the modular characteristic κ at degree two, the classical r -matrix as its binary collision residue, the shadow tower beyond, the genus tower as its trace. The question underneath, stated at the end of the volume: characterise the modular convolution algebra as a geometric object -- proposed answer, the endomorphisms of the Lagrangian embedding $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\text{vac}}(\mathcal{A})$ in a (-2) -shifted symplectic stack of vacua.

2 The objects

Definition 2.1 (Chiral algebra and its three bar complexes). *A chiral algebra on X is a right $\mathcal{D}_X$ -module with chiral product $\mu: j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \rightarrow \Delta_! \mathcal{A}$ satisfying locality, the Borchers identity, and unitality (Beilinson–Drinfeld); augmented when $\varepsilon: \mathcal{A} \rightarrow \omega_X$ splits. Three bar complexes coexist and are never conflated: the Francis–Gaitsgory bar (only the top residue; the Lie shadow); the symmetric bar on $\text{Ran}(X)$ (all operator products, Σ_n -coinvariants); the ordered bar on ordered configurations (all products, no quotient). The averaging map $\text{av}: \overline{B}^{\text{ord}} \rightarrow \overline{B}^{\Sigma}$ is a surjective dg Lie morphism whose kernel carries braid and associator data -- a torsor over $\text{GRT}_1(\mathbb{Q})$ at genus zero, the quasi-modular anomaly of E_2 at genus one. What the symmetric theory forgets, the ordered theory remembers, and the surplus is exactly the R -matrix.*

Definition 2.2 (The five typed objects). *For augmented $\mathcal{A}$: the algebra $\mathcal{A}$; the bar coalgebra $B_X(\mathcal{A})$ (complete conilpotent; the twisting datum, never the bulk); the quadratic coalgebra $\mathcal{A}^i = C_X(s^{-1}V, s^{-2}R)$ of a chosen presentation; the Koszul dual algebra $\mathcal{A}^!$, reached by continuous Ran–Verdier duality $K_X(\mathcal{A}) = \mathbb{D}_{\text{Ran}} B_X(\mathcal{A})$ -- holonomic $\mathcal{D}$ -module duality, never the linear dual; and the derived chiral centre $Z_{\text{ch}}^{\text{der}}(\mathcal{A}) = C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$, the bulk. Bar–cobar inversion $\Omega_X B_X(\mathcal{A}) \simeq \mathcal{A}$ and Koszul duality are different functors with different targets.*

Definition 2.3 (Modular characteristic). *$\kappa(\mathcal{A})$ is the scalar of the genus curvature: $\text{tr}_{\text{diag}}(m_o^{(g)}) = \kappa(\mathcal{A}) \omega_g$ with $\omega_1 = c_1(\mathbb{E})$ the first Hodge class -- equivalently the cyclic trace of the self-sewn binary corolla, equivalently the normalised residue trace of the collision kernel. From the ordered side: $\kappa = \text{av}(r_{\mathcal{A}}(z))$ for abelian families, plus the Sugawara self-contraction $\dim \mathfrak{g}/2$ for nonabelian Kac–Moody. Values: $\kappa(\mathcal{H}_k) = k$ (the level, not $c/2$); $\kappa(V_k(\mathfrak{g})) = \dim \mathfrak{g} (k + h^{\vee})/2h^{\vee}$; $\kappa(\text{Vir}_c) = c/2$; $\kappa(\mathcal{W}_N) = c (H_N - 1)$ with H_N the harmonic number; $\kappa(bc_{\lambda}) = -(6\lambda^2 - 6\lambda + 1)$, $\kappa(\beta\gamma_{\lambda}) = +(6\lambda^2 - 6\lambda + 1)$; $\kappa(V_{\Lambda}) = \text{rank } \Lambda$; $\kappa(V^{\natural}) = 12$.*

3 The theorems

Theorem 3.1 (Selection principle; proved). *For a meromorphic product datum on X , the Fulton–MacPherson bar differential satisfies $d^2 = 0$ if and only if the product satisfies the Borchers identity: disjoint indices contribute Koszul signs, triple collisions contribute the Arnold relation on forms against the Borchers identity on modes, and repeated pairs die because the logarithmic normal factor is consumed by the first residue. The bar complex is synonymous with the chiral-algebra axiom.*

Theorem 3.2 (The adjunction and its honest scope). *The continuous cobar Ω_X is left adjoint to the bar: $\text{Hom}(\Omega_X C, \mathcal{A}) \cong \text{Tw}(C, \mathcal{A}) \cong \text{Hom}(C, \overline{B}_X \mathcal{A})$ -- the counit exists for every augmented chiral algebra, realised by the residue pairing against logarithmic forms, with no Koszul hypothesis. The enhanced statement -- mutually inverse bar and cobar on the pro-nilpotent Francis–Gaitsgory Ran category, transported to the Fulton–MacPherson chain models and closed under factorization -- carries three named*

packages (factorization closure, completed chain comparison, Verdier transport) whose construction for the standard atlas is the open work; the ambient factorization category itself remains to be constructed in full. The honest form: the adjunction backbone and the chain-level geometry are proved; the $(\infty, 2)$ -enhancement is an architecture whose load rests on its packages.

Theorem 3.3 (Koszulness; proved by the PBW criterion). *If $\mathcal{A}$ carries a flat PBW filtration with classically Koszul associated graded and finite-dimensional Mittag-Leffler windows, then the twisted tensor product is acyclic and the quadratic comparison $q_{\mathcal{A}}: \mathcal{A}^i \rightarrow B_X(\mathcal{A})$ is a quasi-isomorphism. Consequences: affine Kac–Moody $\widehat{\mathfrak{g}}_k$ is chirally Koszul at generic level; the Virasoro algebra is chirally Koszul for all c (the quartic pole feeds curvature, not the graded bracket). Seven diagnostics are equivalent on the recognition locus, with the PBW E_2 -collapse of the bar spectral sequence as hub: Koszul twisting morphism; counit and unit equivalences; acyclic twisted products; diagonal bar concentration; diagonal Ext vanishing $\text{Ext}^{p,q}(\mathbf{1}, \mathbf{1}) = 0$, $p \neq q$; Shapovalov non-degeneracy in the bar range. Koszulness is a moduli problem over the associator torsor, and it is GRT_1 -invariant. Off the locus: the critical-level algebra $V_{-h^\vee}(\mathfrak{g})$ is not Koszul (the Feigin–Frenkel centre spreads off-diagonal); the $c_{3,4}$ minimal model is not (a null vector enters the bar range).*

Theorem 3.4 (Curvature and the coderived ambient). *Twisting the curved cobar by α has square $[h_\Omega + d\alpha + \frac{1}{2}[\alpha, \alpha], -]$; genus-zero conilpotent objects live in the ordinary derived category, while genus ≥ 1 with $\kappa \neq 0$ is intrinsically coderived (Positselski’s second kind). The Heisenberg genus-one trace cocycle $\omega_\ell(x_m, y_n) = \ell \, m(x, y) \, \delta_{m+n,0}$ is a nonzero Chevalley–Eilenberg class, and the completed cobar of the finite trace quotient recovers the centrally extended loop algebra: this is the mechanism by which the Koszul dual acquires curvature. The Feigin–Frenkel reflection $k \mapsto -k - 2h^\vee$ is the unique affine-linear involution fixing the critical level and compatible with κ ; class-M chain-level comparisons hold in the pro, adic, or filtered-complete ambients and are provably false on the bounded direct sum – the harmonic discrepancy family $\delta_r^{\text{harm}} = c_r(\mathcal{A}) m_0^{\lfloor r/2 \rfloor - 1}$ admits no single absorbing homotopy.*

Theorem 3.5 (Complementarity; the scalar law). *The centre complex $\mathbf{C}_{g,n}(\mathcal{A}) = R\Gamma(\overline{\mathcal{M}}_{g,n}, \mathcal{Z}(\mathcal{A}))$ splits by the represented Verdier involution into complementary halves $\mathbf{Q}_{g,n}(\mathcal{A}) \oplus \mathbf{Q}_{g,n}(\mathcal{A}^\dagger)$; under the perfectness package the halves are Lagrangian, dual up to the shift $-(3g - 3 + n)$ – what one algebra sees as deformation, its dual sees as obstruction. The scalar shadow is the conductor law*

$$\kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger) = \varkappa : \quad 0 \text{ (free fields, lattices, affine Kac–Moody),} \quad 13 \text{ (Virasoro),} \quad 250/3 \text{ } (\mathcal{W}_3),$$

with $\varkappa = \rho \cdot K$ on the $\mathcal{W}$ -lane, $\rho(\mathfrak{g}) = \sum_i 1/(m_i + 1)$ the anomaly ratio and $K = c + c^\dagger$ the central-charge conductor. The shifted-symplectic upgrade is a separate package.

Theorem 3.6 (The genus tower). *Four objects, four packages. The native class $\text{Obs}_g^{\text{def}} \in H^2(\text{Def}_g(\mathcal{A}))$; its genus-one trace is computed, $\text{tr}_1 \text{Obs}_{1,1}^{\text{def}} = \kappa \lambda_1$; its K -theoretic avatar $\mathfrak{D}_g^K = \kappa \lambda_{-1}(\mathbb{E}_g)$ with Hodge character $(-1)^g \kappa \lambda_g$ is a comparison hypothesis for $g \geq 2$, not a computation; Mumford’s relation $\Lambda_g(t) \Lambda_g(-t) = 1$ gives $\lambda_g^2 = 0$, so the Hodge shadow squares to zero wherever the comparison holds; and the numerical trace is the Faber–Pandharipande integral*

$$F_g = \kappa \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}, \quad \lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g = \frac{2^{2g-1} - 1}{2^{2g-1}} \frac{|B_{2g}|}{(2g)!}, \quad \sum_{g \geq 1} F_g^{\text{sc}} x^{2g} = \kappa(\widehat{A}(ix) - 1),$$

the cross term vanishing for free fields and equal to $(c + 204)/16c$ for $\mathcal{W}_3$ at genus two. Clutching pins the class: $\xi_h^* \lambda_g = \lambda_h \boxtimes \lambda_{g-h}$, $\xi_{\text{irr}}^* \lambda_{-1}(\mathbb{E}_g) = 0$, and a boundary-normalised tautological class agrees with λ_g

on the socle quotient -- on the nose at $g \leq 2$, λ_g -conjecture territory beyond. The honest summary: genus one is a theorem, the higher-genus scalar lane is classical Mumford–Faber–Pandharipande riding a named comparison package, and the non-scalar tower is the open object.

Theorem 3.7 (Hochschild concentration). *The chiral Hochschild complex -- inputs and output on $\overline{C}_{n+2}(X)$ against logarithmic forms -- computes the derived centre, with the first-page spectral sequence $H^q(\overline{C}_{n+2}, \mathcal{A}^{\boxtimes(n+2)} \otimes \Omega_{\log}^n) \Rightarrow \text{ChirHoch}^{n+q}$. Given a family retract onto a model supported in finite $S \subset \mathbb{Z}$, the support is bounded by S ; the anchors are Bakalov–De Sole–Kac: Virasoro support $\{0, 2, 3\}$ with one dimension each, and at the critical level $\text{ChirHoch}^\bullet(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \Lambda(P_1, \dots, P_r) \otimes \mathbb{C}[\Theta_1, \dots, \Theta_r]$, degrees tied to the exponents -- unbounded, the theorem’s own boundary. The force of the theorem for a new family is exactly the construction of its retract; the confrontation with the nonvanishing variational Poisson cohomology of De Sole–Kac is the volume’s sharpest open front.*

Theorem 3.8 (The ordered face). *Descending the ordered bar local system along $\text{Conf}_n^{\text{ord}} \rightarrow \text{Conf}_n$ requires a Σ_n -equivariant structure; the construction of that descent datum is the R -matrix, realised as monodromy of the Kohn connection whose coefficients are bar collision residues. Consequences, each on its stated locus: the ordered Koszul dual of the Heisenberg algebra is the abelian Yangian $\mathbb{C}[h_0, h_1, \dots]$; for the Yang kernel, Verdier duality inverts the R -matrix, giving the self-duality $Y_{\hbar}(\mathfrak{sl}_2)^! \cong Y_{-\hbar}(\mathfrak{sl}_2)$ with $\hbar = 1/(k + h^\vee)$, extended to all classical types on finite windows and reduced, for exceptional types, to a named finite package; genus-one monodromy exponentiates the additive parameter to $q = e^{\pi i \hbar}$ -- a monodromy readout, deliberately not an algebra identification; and Drinfeld–Kohn becomes the statement that Koszul duality reverses collision cycles, $\mathcal{R} \mapsto \mathcal{R}^{-1}$, $q \mapsto q^{-1}$, on the integrable fusion surface. The Francis–Gaitsgory theory is the commutator shadow: a spectral sequence from the Lie bar of the associated graded converges to the graded ordered bar. At the critical level the bar computes opers, $H^n(\overline{B}(\widehat{\mathfrak{g}}_{-h^\vee})) \cong \Omega^n(\text{Op}_{\mathfrak{g}^\vee}(D))$ -- degree zero is Feigin–Frenkel, the higher degrees ride the Frenkel–Teleman comparison -- and the reflection $\kappa(V_k) + \kappa(V_{k^\vee}) = 0$ solves to $k^\vee = -k - 2h^\vee$ -- the Langlands-dual level as a curvature equation.*

Theorem 3.9 (The averaging conductor; proved). *The ordered-to-symmetric averaging av is a natural dg Lie morphism, simultaneously quotient, Reynolds fixed-point summand, and homotopy-coinvariant shadow, lossy exactly on the nontrivial isotypic sector with kernel dimension $d^n - \binom{n+d-1}{d-1}$ in arity n . Its scalar avatars: $K^c = c + c^!$; $K^\kappa = \kappa + \kappa^!$; and the ghost form $K = -c_{\text{ghost}}(\text{BRST})$, whence the conceptual origin of the $\mathcal{W}$ -conductor: $K_N = \sum_{j=2}^N 2(6j^2 - 6j + 1) = 4N^3 - 2N - 2$, one bc -ghost per spin.*

4 The census

The master table (canonical registry: `landscape_census.tex`):

$\mathcal{A}$	Koszul dual	c	$c + c^\dagger$	κ
free fermion ψ	$\text{Sym}^{\text{ch}}(\gamma)$	$1/2$	0	$1/4$
bc_λ	$\beta\gamma_\lambda$	$1 - 3(2\lambda - 1)^2$	0	$c/2$
$\mathcal{H}_k$	curved $\text{Sym}^{\text{ch}}(V^*)$, $m_0 = -k\omega$	1	$--$	k
V_Λ (Λ even unimodular)	self-dual	$\text{rk } \Lambda$	$--$	$\text{rk } \Lambda$
$\widehat{\mathfrak{g}}_k$	$\text{CE}^{\text{ch}}(\widehat{\mathfrak{g}}_{-k-2h^\vee})$	$\frac{k \dim \mathfrak{g}}{k+h^\vee}$	$2 \dim \mathfrak{g}$	$\frac{\dim \mathfrak{g} (k+h^\vee)}{2h^\vee}$
Vir_c	Vir_{26-c}	c	26	$c/2$
$\mathcal{W}_3^k$	$\mathcal{W}_3^{-k-6}$	c	100	$5c/6$
$\mathcal{W}_N^k$	$\mathcal{W}_N^{-k-2N}$	Fateev–Lukyanov	$4N^3 - 2N - 2$	$c(H_N - 1)$
Bershadsky–Polyakov	$\mathcal{B}^{-k-6}$ (conditional)	$-\frac{(2k+3)(3k+1)}{k+3}$	50	open
$N=2$ superconformal	SCA_{6-c}	$\frac{3k}{k+2}$	6	$(k+4)/4$
Monster $V^\natural$	Vir_2 (Virasoro sector)	24	26	12
Yangian $Y_h(\mathfrak{g})$	$Y_{-h}(\mathfrak{g})$	$--$	$--$	residue class

The conductor values: $K_N = 26, 100, 246, 488, 850$ for $N = 2, \dots, 6$, derived three independent ways (ghost spin sums, Feigin–Frenkel screening, Chevalley–Shephard–Todd exponents); $\varkappa = (H_N - 1)K_N = 13, 250/3, 533/2, 9394/15$. Anomaly ratios $\rho(\mathfrak{g}) = \sum_i 1/(m_i + 1)$: $\rho(A_1) = 1/2$, $\rho(G_2) = 2/3$, $\rho(F_4) = 7/8$, $\rho(E_6) = 427/360$, $\rho(E_7) = 2777/2520$, $\rho(E_8) = 121/126$. Bershadsky–Polyakov: $c(k) + c(-k-6) \equiv 50$ proved, self-conjugate value 25 attained only at complex level, κ open, conditional $\varkappa = 25/3$ on the ratio lane (the once-recorded $98/3$ and $K = 196$ are retracted). Self-dual loci are mostly empty -- the fixed point of the parameter reflection is not a categorical self-duality for Heisenberg, affine, or $\beta\gamma$ families; the genuine scalar self-dual point is Virasoro at $c = 13$. The level-one accident: $\kappa(\widehat{\mathfrak{g}}_1) = \text{rank } \mathfrak{g}$ by Frenkel–Kac, the unique cross-class transition ($\mathbb{L} \rightarrow \mathbb{G}$); E_8 level one is the unique Koszul self-dual lattice theory. The corner-algebra witness that κ is not $c/2$: $Y_{1,1,1}[\Psi]$ has $c = 0$ and $\kappa = \Psi$.

Shadow data: $S_2 = \kappa, S_3, S_4$ with discriminant $\Delta = 8\kappa S_4$; the weighted tower is algebraic of degree two, $H(t)^2 = t^4 Q(t)$, which forces the depth dichotomy $r_{\max} \in \{2, 3, 4\}$ or ∞ and the five classes $\mathbb{G}/\mathbb{L}/\mathbb{C}/\mathbb{M}/\mathbb{FF}$ (Gaussian, Lie, contact, mixed, critical). Virasoro: $S_3 = 2, S_4 = 10/[c(5c + 22)]$, $\langle \Lambda | \Lambda \rangle = c(5c + 22)/10$, growth radius $|c|/6$.

The arithmetic shadow, proved core first. The Verlinde volumes $Z_g(k) = \left(\frac{n}{2}\right)^{g-1} \sum_j \csc^{2g-2}(\pi j/n)$ are polynomials whose leading coefficients are Bernoulli, $(-1)^g 2^{g-1} B_{2g-2}/(2g-2)!$, and the Kummer-irregular primes 691, 3617 appear at genus 7 and 9 exactly where the Bernoulli numerators put them -- while the Virasoro shadow numerators through $r = 11$ contain none of them: a sharp arithmetic duality between the two localisations, with its near-counterexample (2111) recorded. The Virasoro tower is motivically pure: algebraicity of the weighted generating function over $\mathbb{Q}(c)$ kills all multiple zeta values -- depth one is rational, depth climbs only with the operadic level (E_2 receives $\zeta(n)$ through the associator, E_3 depth two). For lattice theories the shadow–spectral dictionary holds: theta decomposition against Eisenstein and Hecke forms factorises the constrained Epstein zeta, with the Leech witness $S_{12} = -65520/691$ -- Ramanujan’s congruence surfacing as a shadow coefficient. The shadow L -function $L^{\text{sh}}(\mathcal{A}; s) = \sum_r S_r r^{-s}$ is formal: the proved statement is the genus-slot bridge, deliberately

not a functional equation, and the volume's own retraction of the Riemann-style evaluation is part of the record.

5 The garden

One logarithmic 1-form generates the programme; this volume is where the generation happens, so the map here is outward. Every object sits on one chain: primitive factorization category $\rightarrow$ chart algebra $A_b \rightarrow$ bar shadow $\overline{B}(A_b) \rightarrow$ derived chiral centre $Z_{\text{ch}}^{\text{der}}(A_b)$ (the bulk) $\rightarrow$ acting line operators $\rightarrow$ scalar modular shadow. Identifying stages without the licensing datum -- shadow = object -- is the master error the programme is disciplined against.

Volume II (chiral-bar-cobar-vol2) reads this volume's engine as physics: bar = BV/BRST, quantum master equation = Maurer–Cartan, the anomaly identity $Q_{\text{BRST}}^2 = \frac{c-26}{12}c_o \leftrightarrow d_{\text{fib}}^2 = \kappa_{\text{tot}}\omega_1$ with $\kappa_{\text{tot}} = (c - 26)/2$, the two-coloured Swiss-cheese operad as bulk–boundary home, and Universal Holography $\mathcal{A} \mapsto (\partial = \mathcal{A}, \text{bulk} = Z_{\text{ch}}^{\text{der}}(\mathcal{A}))$ -- at $\text{Vir}_{c=3\ell/2G_N}$ the boundary reading of 3d gravity, with the same conductor 26 appearing as the critical dimension. *Volume III* (calabi-yau-quantum-groups) constructs the geometric source: the two-stage functor $\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}}$ from Calabi–Yau categories to chiral algebras, the cohomological Hall positive half and its Drinfeld double as the level-three quantum group, the κ -tuple $(0, 0, 3, 5, 24)$ at $K3 \times E$. *Mixed holomorphic-topological strings* computes the local germ of the two-dimensional theory at N Dirac branes -- the trace coordinate $J(f) = \text{Tr} f(\phi_1, \phi_2)$ with Capelli anomaly -- and the typed ledger separating germ from globe. *The Igusa cusp form* (igusa-cusp-form) is the terminal scalar: Δ_5 read five ways, $Z_{\text{BPS}}^{K3 \times E} = \Delta_5^{-2}$, the order-two orientation bit that squaring kills -- the sharpest single example of this volume's slogan that the scalar shadow underdetermines the object.

The single question underneath: characterise $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ geometrically. This volume computes its Taylor jets -- the shadow tower -- and proposes the answer; Volume II supplies the local composition law, Volume III the geometric source, the satellites the terminal example.

6 The frontier

The honest list, in the volume's own post-audit order:

- (i) The ambient factorization category for the full $(\infty, 2)$ -form of the adjunction -- until constructed, the enhanced Theorem A reads: where such a category exists, bar–cobar and Positselski apply in it.
- (ii) The Hochschild engine: ordered twisted-tensor acyclicity, and the confrontation with De Sole–Kac nonvanishing -- the single most important unresolved mathematical question of the volume.
- (iii) The genus tower beyond the scalar-diagonal lane: the identification $\text{Obs}_g^{\text{def}} = \kappa\lambda_{-1}(\mathbb{E}_g)$ for $g \geq 2$ as computation rather than package; the non-scalar Maurer–Cartan obstruction, governed by Bianchi, not killed by Mumford.
- (iv) Class M on the bounded direct sum -- provably false there; the completed ambients are the theorem, the descent is the question.
- (v) Fingerprint completeness: the five-slot invariant is a proved projection; the reconstruction direction rides a package that essentially assumes it.
- (vi) The arithmetic superstructure: the uniform Kummer-absence threshold; the noncommutative Weil dictionary $(\Theta_{\mathcal{A}} \leftrightarrow \text{Frobenius}, \text{positivity} \Rightarrow \text{Ramanujan})$ -- a suggestive dictionary whose

mechanism is open, with Ramanujan for lattices imported from Deligne; the K_3 ladder of recognition targets (tree Δ_5 , one-loop Heegner, three-loop $\zeta(3)$, $25/3$, $\Phi_{10}^{\text{un}}/\eta^{24}$), every rung conjectural.
(vii) Bershadsky–Polyakov’s genus-one curvature -- the one census row still open.

What stands without hypothesis: the selection principle; the adjunction backbone; the ordered bar functor and its shuffle theorem; the PBW recognition of Koszulness with its Kac–Moody and Virasoro instances; the curvature mechanism and the central-extension extraction; the completion theory; the averaging conductor; the census with its three-way conductor derivations; genus one of the tower; the Verlinde–Bernoulli arithmetic and the motivic purity of the Virasoro shadow. The volume’s deepest discipline is its own maxim -- primitive objects first, shadows second, scalar modular forms last -- and the theorems above are exactly the places where a shadow was promoted to an object by an honest construction.